



Mehran University of Engineering & Technology
Department of Computer Systems Engineering

21CS 4th Semester EEPM Final Project

Group Members:

Kamran Ali (21CS055)

Muhammad Sameer (21CS077)

Mashaal Zulfikar (21CS105)

Section: 21CS-1

Submitted to: Sir Muhammad Ali Khan

Date: 10/07/2023





Case Study: Investigating the Causes of Downfall of Finance & IT Organizations

Kamran Ali (21CS055)

Muhammad Sameer (21CS077)

Mashaal Zulfikar (21CS105)

Table of Contents

[Table of Contents](#)

[Introduction](#)

[Chapter 1 - The Big Short: How Greed in Wall Street Banks caused 2000s US Housing Market Collapse](#)

[What is *The Big Short*?](#)

[Mortgage Backed Security](#)

[How MBS Works](#)

[Understanding Subprime](#)

[The Big Short Jenga Blocks](#)

[Dr. Michael Burry](#)

[Shorting the US Housing Market](#)

[The Rise and Fall of Lehman Brothers: A Financial Journey](#)

[The Deutsche Bank Debacle: A Tale of Financial Turmoil](#)

[Chapter 2 - Mistakes & Failures of Tech giants: Yahoo and Nokia](#)

[The Rise, Fall, and Resurgence of Nokia: A Tale of Adversity and Redemption](#)

[Reasons for Downfall of Yahoo!](#)

[Dictionary](#)

[References](#)

Introduction

“It ain't what you don't know that gets you into trouble. It's what you know for sure that just ain't so.” ~ Mark Twain

In this article, we're going to dig deep into why some big finance and IT companies have failed over time. These are the giants of their industries, but they've faced big challenges and even collapses. By looking into these cases, we can figure out why these once-successful companies went down.

Why is this important? Because it gives us important lessons on how to prevent similar failures in the future, protect people's money, and keep our economy stable. And, we'll also connect what we learn to our course Engineering Economics & Project Management. We'll use real-world examples to show how understanding risk, managing money wisely, and handling unexpected challenges are super important in our course and in the real world.

Our journey has two key chapters. First, we'll dive into "The Big Short: How Greed in Wall Street Banks caused the 2000s US Housing Market Collapse." This is a story of how some big banks got greedy, made risky bets, and caused a huge housing

crisis in the United States.

Then, in our second chapter, we'll look at the tech world. We'll explore the "Mistakes & Failures" of companies like Yahoo, and Nokia. They were once giants in their field, but they made big mistakes and lost their top spots. We'll learn what went wrong and how they could have done things differently.

As we go through these chapters, remember that the lessons here aren't just for big companies; they're for everyone. We'll see why it's important to think about risks, manage money wisely, be ethical in our decisions, follow the rules, and be ready to handle unexpected challenges.

Join us on this journey as we uncover why these big companies fell and what we can learn from their stories. It's not just about their past mistakes; it's about making a better future for ourselves and our projects in Engineering Economics & Project Management.

Chapter 1 - The Big Short: How Greed in Wall Street Banks caused 2000s US Housing Market Collapse

Before mentioning the Wall Street Banks and their mistakes that caused the the 2000s US Housing Market Collapse, We should first understand *The Big Short*.

What is *The Big Short*?

The Big Short is a course of events that led to "Subprime Mortgage Meltdown". Back in the late 70s in the US, banking was boring. Its sole purpose was accounting or selling insurances. Banking wasn't such a profitable industry for the US economy until Lewis Ranieri, a bond trader at Salomon Brothers and a prominent figure in finance industry, pitched the idea of "Mortgage Backed Security" (MBS).(1)



Lewie Ranieri is known as the father of the mortgage-backed security. Now, he's crusading to rescue the real estate market that his invention helped destroy.(2) - December 09, 2009

Mortgage Backed Security

A Mortgage Backed Security (MBS) is a financial instrument that represents a claim on the cash flows generated by a pool of mortgage loans. These loans are typically home mortgages. MBS effectively bundle numerous individual mortgages into a single security, which is then sold to investors. The investor who buys a mortgage-backed security is essentially lending money to home buyers. An MBS can be bought and sold through a broker. The minimum investment varies between issuers.(3)

How MBS Works

MBS works by aggregating thousands of individual mortgage loans, which may have varying interest rates, terms, and risk profiles, into a single investment product. Investors in MBS receive periodic payments that come from the homeowners' mortgage payments. These payments include both the interest on the loans and the return of the principal amount borrowed.

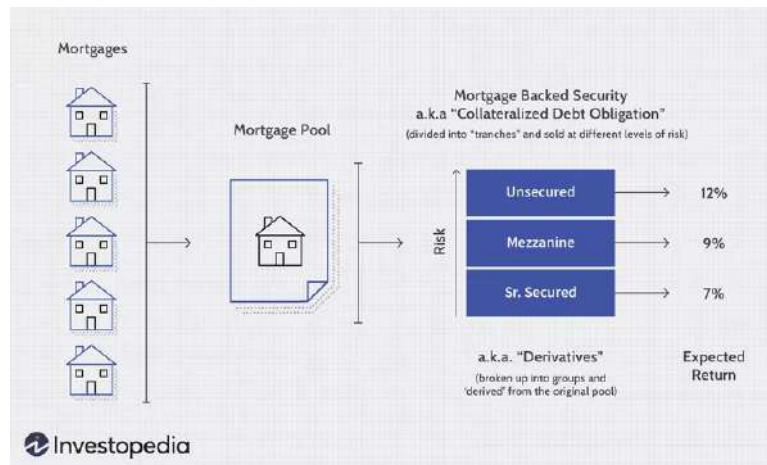


Image by Sabrina Jiang © Investopedia 2020 (4)

Understanding Subprime

Investors were desperate to buy more and more Mortgage Backed Securities. So, lenders (banks) did their best to help create more of them. But to create more of them, they needed more mortgages. So lenders loosened their standards and made loans to people with low income and poor credit. You'll hear these called "Subprime Mortgages".

Eventually, some institutions even started using what are called predatory ending practices to generate mortgages. They made loans without verifying income and offered absurd, adjustable rate mortgages with payments people could afford at first, but quickly ballooned beyond their means.

But these new subprime lending practices were brand new. That meant credit agencies (e.g. U.S. Securities and Exchange Commission SEC) could still point to historical data that indicated mortgage debt was a safe bet. But it wasn't. These investments were becoming less and less safe all the time. But investors trusted the ratings, and kept pouring in their money. Traders also started selling an even riskier product, called collateralized debt obligations, or CDOs. And again, some of these investments were given the highest credit ratings from the ratings agencies, even though many of them were made up of these incredibly risky loans. While, the investors and traders and bankers were throwing money into the U.S. housing market, the U.S. price of homes was going up and up and up and up.(5)

The Big Short Jenga Blocks

In the following image of an iconic Jenga Tower, we encounter a towering structure, much like a Jenga block, representing a complex financial concept - the mortgage-backed security. Picture it like this: at the pinnacle of this towering financial structure, we have the AAA-rated mortgages, considered safe and virtually risk-free. They form the stable upper levels of our Jenga tower.

However, as we descend towards the base, things take a riskier turn. Here, we encounter the likes of BBB, BB, and even lower-rated mortgages or much higher probability of "default mortgages", considered much riskier, even borderline junk. These occupy the lower sections of our financial Jenga.

Now, here's where it gets interesting. If you begin removing Jenga pieces from the bottom, representing those lower-rated B mortgages that default, and then another piece goes, and another - the tower, our proxy for the U.S. housing market, starts swaying.

As you keep extracting those pieces, they inch their way up towards the middle, what's known as the "mezzanine layer." And if, by some twist of fate, even the AAA, AA, and A-rated mortgages start to default, that's when our financial Jenga tower comes crashing down. It's a vivid representation of the U.S. housing market's collapse.(6)



A sample of a Jenga Tower, representing itself as a Mortgage-Backed Security, and its blocks represent the tranches in MBS.

Dr. Michael Burry

Dr. Michael Burry was a physician turned hedge fund manager who founded Scion Capital. In November 1996, during his residency in neurology at Stanford Hospital, he started a discussion group on "value investments" which he later made into a popular blog. His forecasts were amazingly accurate. He was recognizing opportunities no one else was seeing. In November 2000, Burry quit neurology and started his own hedge fund, Scion Capital, funded by a small inheritance and small loans from his mother and three brothers. Soon enough his fund raised several million dollars from some big funds and a few private investors.

Scion quickly earned exceptional profits for its investors. In 2001, while the S&P 500 fell 13%, Scion was up by 55%. Burry was able to achieve these returns by shorting overvalued tech stocks at the peak of the internet bubble. In 2002, the S&P 500 fell by 23.4%, and Scion was up by 16%. In 2003, the stock market turned around and rose 26.4% while Scion rose by 50 percent.



Dr. Michael Burry, who predicted 2008 financial crisis bets \$1.6bn on stock market crash by end of 2023. Security Exchange Commission filings released on Monday show that he has taken out negative options on the S&P 500 and the Nasdaq 100 – both of which are representative of the US economy at large.(7)

Shorting the US Housing Market

From 2003, Burry closely followed U.S. credit instruments. In 2005 he realized that the subprime mortgages, especially those with adjustable interest rates, and the bonds backed by these mortgages, might collapse as early as 2007, when the interest rates would become much higher. He decided to short the market by acquiring insurance against the collapse of specific mortgage backed bonds. In May 2005, he made his first deal, purchasing from Deutsche Bank \$60 million of what became known as credit default swaps (CDS) against six mortgage bonds he had chosen. Later he bought additional CDSs from other firms, such as Goldman Sachs and the Bank of America. By the end of July 2005, his stock of CDSs stood at \$750 million. In October 2005, he sent e-mails to his investors telling them that they owned more than \$1 billion in CDS.(8)

Burry was probably the first fund manager to invest in credit default swaps on subprime mortgage bonds. Being a pioneer in this venture caused him much trouble, since Scion Capital had to pay premiums for nearly three years before seeing profits from the collapse of the mortgage market. When some of his investors demanded to withdraw their investments, Burry utilized a paragraph in the agreement with his investors which allowed him, under certain circumstances, to hold their money. In January 2007, Burry had to explain to his investors why he was not releasing their money and why, in 2006, when the S&P 500 climbed by more than 13%, Scion lost 18.4% due to the CDS premiums it had to pay. Some big investors were furious and threatened to

sue him. However, in February 2007, Burry's analysis proved correct: subprime loans were defaulting in record numbers and financial institutions began to shake. In the first quarter of 2007, Scion rose nearly 18% and the climbing continued. In April 2008, Burry liquidated his CDSs and then announced the shutting down of the fund. His personal profit was \$100 million and the profit for his investors was more than \$700 million. Scion Capital net returns since its launch in November 2000 until its liquidation in April 2008 were almost 490%.⁽⁹⁾

Dr. Burry's foresight and his decision to short the bonds would prove to be prescient, as it set in motion a series of events that would ultimately lead to the financial crisis of 2008. His narrative serves as a testament to the importance of careful financial analysis and risk assessment in the world of finance.

The Rise and Fall of Lehman Brothers: A Financial Journey

Lehman Brothers, a name that once echoed through the canyons of Wall Street, holds a special place in the world of finance. This is not just the story of a company; it's a tale of ambition, risks, and the rollercoaster ride of high finance.



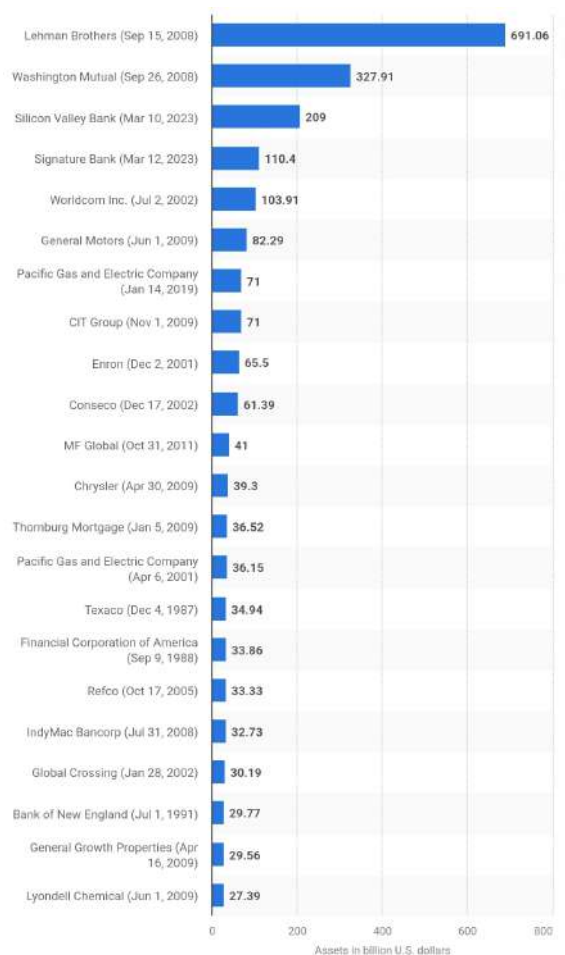
On September 15, 2008 investment banking giant Lehman Brothers filed for Chapter 11 bankruptcy, becoming the largest bankruptcy by asset value.
(10)



Lehman Brothers headquarters in Times Square, a year before the company's collapse.

The Financial Goliath

At its peak, Lehman Brothers was a giant, with assets worth a jaw-dropping \$691 billion. To give you a sense of how big that is, it's larger than any other company that ever went bankrupt in the U.S. It wasn't just any company; it was like a financial titan.



Largest bankruptcies in the United States as of July 2023, by assets at time of bankruptcy. Lehman Brothers had assets worth 691 billion U.S. dollars when filing for bankruptcy on September 15, 2008.(11)

The Humble Beginnings

The story of Lehman Brothers starts in a small dry goods store in 1850s Alabama. It was founded by Henry Lehman, who came from Germany. He began as a door-to-door salesman until he saved enough to start a little store. Henry's two brothers, May and Emanuel, joined him in the business, and that's when Lehman Brothers officially began. It's pretty amazing that this financial giant started from such a modest place.

From Cotton to Wall Street

But it wasn't Wall Street right away. Their journey took an interesting turn when they got into the cotton business. In those days, cotton was like money, especially in the American South. People would buy things with cotton instead of cash. The Lehman Brothers saw an opportunity to be the go-between, connecting cotton farmers with cotton buyers and making some money in the process. This was their ticket to getting into finance.(12)

Adapting to Change

The brothers were quick to adapt to changes in the market. They started helping companies go public, meaning they helped companies sell their shares to the public. This is like connecting a company to people who want to invest in it. They also helped with mergers and acquisitions. This made them really important in the business world. They were good at adapting and staying ahead of the game.(13)

The Acquisitions and Transformations

In the 1980s, Lehman Brothers became part of American Express. After a while, they became their own independent company in 1994. Over the years, they bought other companies, including ones that dealt with subprime mortgages. This showed their ability to change and grow with the times.

**SHEARSON
LEHMAN
BROTHERS**



Shearson/American Express, an American Express-owned securities company focused on brokerage rather than investment banking, acquired Lehman in 1984, for \$360 million. On May 11, the combined firms became Shearson Lehman/American Express.(14)

The Perfect Storm

Now, the tough part. The subprime mortgage crisis hit, which led to people not being able to pay their mortgages. Lehman Brothers had a lot of investments tied to these mortgages, and when the values of these investments started falling, they faced huge losses. It was like a storm hitting a ship with no way to weather it. On September 15, 2008, they filed for bankruptcy.

Figure no.1: Evolution of Lehman Brothers stock during June 2004 – October 2010



Evolution of Lehman Brothers stock during June 2004 – October 2010(15)

The Legacy

Lehman Brothers' bankruptcy had big effects. The stock market tumbled, and it set off a bigger financial crisis. It led to a global recession, which affected economies all over the world. Lehman Brothers, once a symbol of financial success, ended up a symbol of how even the biggest companies can face big troubles.

In Conclusion

In the end, Lehman Brothers' story reminds us that even the mightiest financial giants can fall. They went from a small store in Alabama to the top of the finance world, and then they fell. Lehman Brothers' rise and fall is a lesson in how finance can be a wild ride, and it's a reminder that in the world of money, no one is invincible.

The Deutsche Bank Debacle: A Tale of Financial Turmoil

Deutsche Bank, a name that once resonated as one of Europe's financial giants, has undergone a dramatic transformation over the years. From prosperity to perils, the bank's journey is one that exemplifies the volatile world of finance.(16) Let's dive into the pages of its tumultuous history.



Deutsche Bank Twin Towers in Frankfurt, Germany.

The Price Plunge

If we were to sketch Deutsche Bank's stock price trajectory, it would resemble a rollercoaster ride. Following its recovery from the 2008 financial crisis, the bank saw a staggering 95 percent drop in its market value. This was just the beginning of a series of setbacks that would define its recent history.



Deutsche Bank's Market Share Trajectory: It fell to a new historic low of €5.81 on June 3, and on Friday closed at €6.03. Its shares have now plunged 95% from the peak in 2007.(17)

The Weight of Fines

Deutsche Bank's troubles began to mount with substantial fines. Over the past decade, it has shelled out a staggering \$18 billion in penalties. More than \$10 billion of this amount was incurred in the last five years. In 2015, the bank was slapped with a \$2.5 billion fine by US and UK regulators for manipulating LIBOR.(18) In that same year, it faced another fine of \$250 million for violating US sanctions against Iran and Syria.(19)

The Mortgage Misadventure

In 2008, Deutsche Bank was a major player in the world of mortgage-backed securities. Not only was it selling these investments, but it was also betting against them. This high-risk strategy came crashing down in September 2016 when the bank was hit with a jaw-dropping \$14 billion fine for the improper sale of mortgage securities in the US. The final settlement with the US Department of Justice came in at \$7.2 billion.(20)

The Money Laundering Web

Deutsche Bank's involvement in a money-laundering scheme called the "global laundromat" further muddled its waters. This scheme allowed Russian criminals to move \$80 billion, with \$10 billion of that amount eventually traced to Deutsche Bank. This resulted in a hefty \$630 million fine in 2017.(21)

Fines and Fumbles

Deutsche Bank's path was paved with fines, each one seemingly more substantial than the last. The fines accumulated, leading to a tumultuous period for the bank. Some important fines and penalties raised over Deutsche Bank's violations are listed below:(22)

S.No.	Fine Year	Fine Amount	Fine Description
1	2008	\$3.3 Billion	Deutsche Bank's investors' protection violation due to 2008 global financial crisis
2	2013	\$1.9 Billion	Deutsche Bank's toxic securities abuses
3	2015	\$2.5 Billion	US & UK regulators fine Deutsche Bank for manipulating LIBOR
4	2015	\$250 Million	Deutsche Bank violates US Sanctions against Iran and Syria
5	September 2016	\$7.2 Billion	Improper sale of mortgage securities in the US by Deutsche Bank
6	2017	\$630 Million	Deutsche Bank's "global laundromat" scheme allowed illegal and subsequent advantages to Russian cartels
7	2017	\$95 Million	Deutsche Bank's ax violations

Ties with Trump

The bank's association with Donald Trump added yet another layer of complexity to its situation. Despite Trump's company defaulting on a \$640 million loan, Deutsche Bank continued to lend him money. A detailed investigative report also revealed that some of Deutsche Bank's loans to Trump were underwritten by subsidiaries owned by Russian VTB Bank.(23)

Losing Investor Trust

Deutsche Bank's troubles were reflected in its stock price, trading at a fraction of its book value. The European Central Bank's policy of negative interest rates further impacted its profitability, as the bank effectively paid for its reserves.

A Glimmer of Hope

Despite the immense fines and challenges, Deutsche Bank still stands. Investors have poured over €29 billion into the bank since 2010. Some believe that, aside from the one-time fines, the bank remains solvent and close to profitability.

A Turnaround Attempt

Deutsche Bank is determined to steer itself out of the storm. It's looking to separate unprofitable divisions and focus on core areas. In 2019, the bank planned to establish a "bad bank" to hold toxic assets, but regulations demanded more capital against derivatives, making it a costly endeavor.

The Government's Intervention

The German government supported a merger between Deutsche Bank and Commerzbank in an effort to create a strong national bank. However, the risks and costs outweighed the potential benefits, leading to the deal's demise.

A Grueling Path Forward

Deutsche Bank's latest cost-cutting plan, unveiled in 2019, aims to reduce expenses by €6 billion. This includes downsizing the workforce by 20% and exiting the global equities and trading business by 2022.(24)

The Uncertain Horizon

The question now is, where does Deutsche Bank go from here? Can it resurrect itself from the depths of financial adversity, or will it follow in the footsteps of other fallen giants like Lehman Brothers? Only time will tell if this financial titan can rise again.

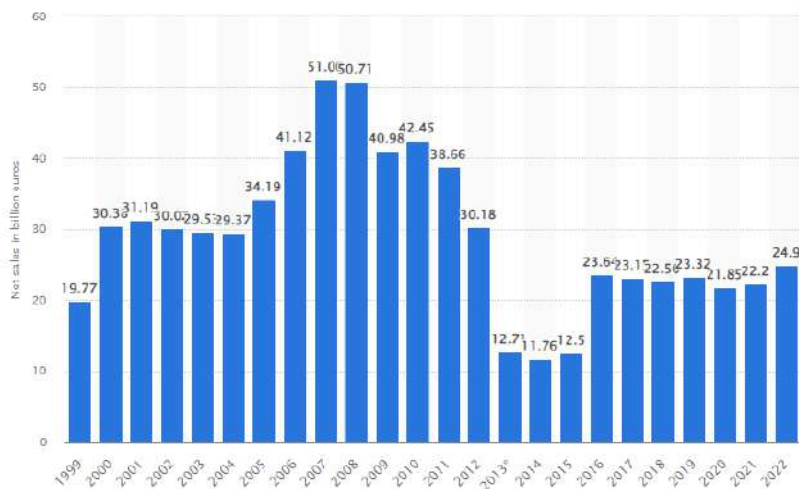
Chapter 2 - Mistakes & Failures of Tech giants: Yahoo and Nokia

In the ever-evolving landscape of technology, even giants can falter. In this chapter, we venture into the world of two tech titans, Yahoo and Nokia, whose stories are emblematic of the volatile nature of the industry. These two companies once stood at the zenith of their respective domains, shaping the digital universe in ways that were, at one point, seemingly unstoppable.

However, as time passed and technology advanced, these giants found themselves struggling to keep pace with the relentless march of progress. Their stories serve as cautionary tales for any company navigating the turbulent waters of the tech world. Join us as we dissect the missteps, missed opportunities, and the hubris that led to the downfall of Yahoo and Nokia, and discover the invaluable lessons their experiences offer in a rapidly changing technological landscape.

The Rise, Fall, and Resurgence of Nokia: A Tale of Adversity and Redemption

Once upon a time, there was a melody that everyone recognized. The iconic Nokia ringtone was the anthem of the early 2000s, signifying the dawn of the mobile revolution. Nokia stood as the undisputed king of cellphones, commanding over 50% of the global mobile market and selling more than 100 million phones annually.(25) Not only were they a mobile giant, but their market capitalization reached nearly \$300 billion at its zenith. However, as the winds of technological change blew, Nokia's tune changed. The rise of smartphones dealt a heavy blow, plummeting their annual sales from a peak of over 100 million to under 20 million phones per year, with a catastrophic decline in annual revenue from \$60 billion to a mere \$6 billion.(26)



Nokia net sales worldwide from 1999 to 2022.(27)



Revenue history for Nokia from 2001 to 2023.(28)

Amid this downward spiral, Nokia was left grappling with a staggering annual loss of \$6 billion, where their expenses far exceeded their revenue. A lifeline arrived when Microsoft swooped in to purchase Nokia's phone business for \$7.2 billion, but it would later become a transaction synonymous with failure, ending in Microsoft writing off the entire acquisition value and selling off the remaining parts for a paltry \$350 million. However, Nokia's story didn't end there. Against all odds, Nokia persisted and re-emerged as a contender, with surprising success, re-entering the smartphone market and posting impressive sales figures.

Join us on this journey to explore how Nokia navigated the treacherous waters of the smartphone era and not only survived but thrived, all after hitting rock bottom.

NOKIA



Nokia's head office until April 2014. The building is located by the Gulf of Finland in Keilaniemi, Espoo, and was constructed between 1995 and 1997. It was the workplace of more than 1,000 Nokia employees.

The Fall Begins: 2007 - The Apex Year

In 2007, while the world was witnessing the grandeur of the iPhone's launch, Nokia was reigning supreme. But Nokia had its foot in the smartphone race even before Apple's iconic device hit the market. They introduced a Linux-based mobile operating system called Maemo in 2005, and touch screen phones in 2008. The trouble lay in Nokia's neglect of the shift in consumer preferences toward functionality over durability. As phones evolved into pocket-sized computers, Nokia clung to its outdated Symbian operating system, isolated from developer support. Meanwhile, competitors like Apple and Android gained momentum.

Nokia's oversight extended to hardware as well. They failed to upgrade internal components and lagged behind in the race for better specifications. While others embraced 3G and 4G technology, Nokia clung to 2G. Their branding, or the lack thereof, was another pitfall. Nokia failed to build an appealing identity for its smartphones, relying on convoluted naming conventions that left customers baffled.(29)

Overconfidence was a looming menace. Nokia underestimated the value of its brand, assuming it could carry them through the smartphone revolution. But consumers saw Nokia as a legacy brand associated with basic cellphones, not cutting-edge smartphones. Their overconfidence, failure to address software and hardware issues, and inadequate branding all played a role in their downfall.

Microsoft Steps In: A Hail Mary with a Hiccup

The predicament faced by Nokia was shared by Microsoft, who had missed the smartphone revolution and was desperate to stage a comeback. In a bold move, Nokia's then-CEO, Stephen Elop, invited Microsoft into the fold. This collaboration would go down as a critical juncture, but not for the better.(30)



Steve Ballmer and Stephen Elop at a Microsoft-Nokia event in New York in September 2012.

Microsoft's operating system, Windows Mobile, proved ill-suited for the touch screen era. It failed to garner developer support, making Nokia's smartphone experience lackluster. Hardware problems persisted, and Nokia's smartphones couldn't match their competitors in terms of specifications.

Elop's maneuver seemed like a last-ditch effort to save Nokia, but it mainly sealed their fate. Under Elop's leadership, Nokia's stock price plummeted, mobile phone market share dwindled, and their smartphone market share fell from 33% to a mere 3%. The company incurred a loss of €4.9 billion.

But the most concerning aspect of this phase was Elop's apparent self-preservation at the expense of the company. Instead of turning to Android, he forced Microsoft into acquiring Nokia's phone business. This acquisition turned out to be a complete disaster, with Microsoft later abandoning the phone business.(31)

Nokia's Resurgence: Reinvention and Redemption

The saving grace for Nokia was that Microsoft did not acquire their entire business. They only acquired the phone segment, leaving Nokia with their network business. While this seemed like a grim prospect initially, it paved the way for Nokia's reinvention. Faced with the challenge of survival, Nokia directed its efforts towards revamping its network business, effectively addressing its previous blunders.

Nokia embraced the networking equipment sector and distanced itself from its branding, recognizing that enterprise customers prioritized results over the name. Their overconfidence disappeared as they fought for survival, leading to a humbler and more pragmatic approach.

In the absence of their phone business, Nokia found solace in networking equipment. They evolved and repaired their past mistakes,

setting an industry standard for 5G-ready networks in 2016.

Moreover, Nokia made a triumphant return to the smartphone market, partnering with Google and unveiling the first Nokia Android smartphone in 2017. Although they might never regain their former glory, Nokia successfully re-entered the smartphone market and has been selling millions of phones annually. A once-forgotten brand had turned adversity into a lesson in resilience. (32)

Conclusion

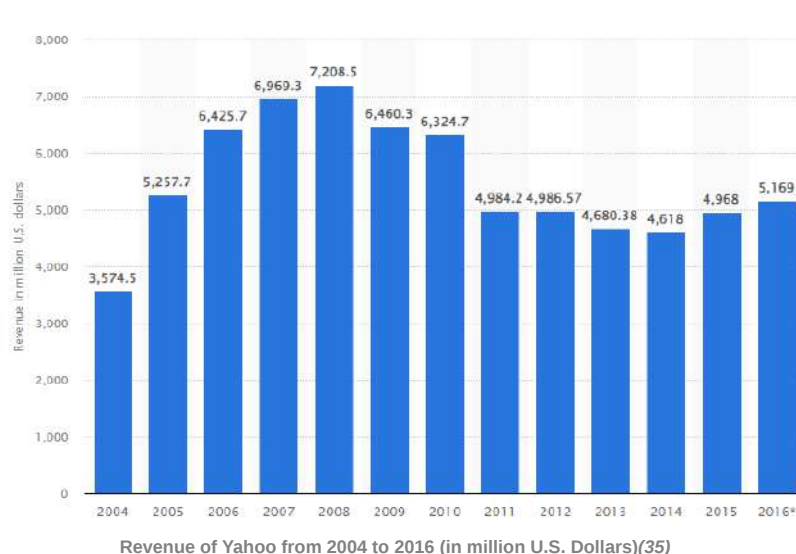
The story of Nokia is a tale of transformation, resilience, and redemption. Despite teetering on the brink of irrelevance after their failure with Microsoft, they reinvented themselves by focusing on networking equipment and making a successful return to the smartphone market. Nokia's journey serves as a testament to the power of adaptation and the importance of learning from past mistakes in a rapidly evolving technological landscape.

Reasons for Downfall of Yahoo!



Yahoo! headquarters in Sunnyvale, California.

Yahoo! – once considered as an internet giant is now struggling for its existence. Yahoo Inc., an American multinational technology company headquartered in Sunnyvale, California was founded by Jerry Yang and David Filo in January 1994 and was incorporated on March 2, 1995.(33) With more than 7 billion views per month, Yahoo is still ranked as the fifth most viewed website around the world by Alexa, a leading third-party web analytics provider.(34) Yahoo is globally known for its Web portal, search engine Yahoo! Search, and related services, including Yahoo! Directory, Yahoo! Mail, Yahoo! News, Yahoo! Finance, Yahoo! Groups, Yahoo! Answers, advertising, online mapping, video sharing, fantasy sports, and its social media website. However, the company which was a pioneer of early internet in 1990s and early 21st century, with so many operations and once valued over \$100 billion, is currently negotiating its sale to Verizon, an American telecommunication giant for a sum of just \$4.8 billion which is almost 10 times below the company's current market value, we will discuss the reasons for the downfall of such a giant in details



1. Failure to Innovate:

Yahoo was initially a pioneer in the early days of the internet, offering services like web search, email, and news. However, it struggled to innovate and keep pace with changing technology and user preferences. It missed the opportunity to develop its own search engine technology and instead relied on third-party solutions, like Microsoft's Bing, which limited its competitiveness in search.(36)

2. Competition:

Google emerged as a dominant force in the search industry, offering a superior search engine and rapidly expanding into other areas like advertising and email. Yahoo faced intense competition from Google, which led to a decline in its search market share and advertising revenue. Additionally, competitors like Facebook and Twitter gained prominence in the social media space, drawing users and advertisers away from Yahoo!.

3. Acquisition Failures:

Yahoo! invested a lot of money in acquiring companies like Broadcast.com, Geocities, Flickr, Right media, Overture Services, Inc, and Tumblr, all decisions leading it one step down the hill.

3.1. Failure with Tumblr

Yahoo purchased the young Tumblr micro blogging organization in 2013 for \$1.1 billion, but failed to turn their acquisition into a profitable component of the company. After investing hundreds of millions of dollars into the project, Tumblr essentially disappeared into obscurity and proved to be one of the last opportunities for Yahoo to redeem itself in the business world. (37)

3.2. Failure with Flickr

Before Yahoo purchased Flickr in 2005, its founders had intended to tap into social networking. However, the plan wasn't carried forward, and they missed out on the social media boat. There was a lack of vision, and it paid less attention to cater to the community it had built. It could not retain its customers or expand the customer base because the focus was not on introducing new features or growing the community.(38)

4. Monetization Challenges:

Yahoo's primary source of revenue was online advertising, but it faced challenges in monetizing its large user base effectively the rise of more targeted and efficient advertising platforms, particularly Google Ads, made it difficult for Yahoo to compete and attract advertisers.

5. Security Breaches:

On September 22, 2016, Yahoo made an official announcement that "A recent investigation by Yahoo has confirmed that a copy of certain user account information was stolen from the company's network in late 2014 by what it believes is a state-sponsored actor. The account information may have included names, email addresses, telephone numbers, and dates of birth, hashed passwords and, in some cases, encrypted or unencrypted security questions and answers." The statement also revealed that "information associated with at least 500 million user accounts was stolen "The Company had also advised its users in this statement to change their login passwords and security questions and answers.(39)

6. Lack of Clear Identity:

From the beginning, Yahoo as a whole lacked a clear vision regarding the overall purpose of the company. They were not a search company, nor a tech company, but functioned primarily as a media company that considered programming to be a necessary commodity to translate production work into code. Coupled with a string of poor leaders lacking vision for the future, Yahoo's ultimate demise was foreseen years beforehand by some former employees.(40)

7. A revolving door of CEOs

From the late 1990s through the 2000s and into the 2010s, Yahoo experienced a constant churn of chief executives, each ushering in their vision and strategy. The cycle of changing CEOs began in the early 2000s when Yahoo co-founders Jerry Yang and David Filo handed over leadership to a series of external hires. The lack of continuity in leadership meant that the company lacked a clear, long-term vision and strategic direction. Each new CEO came with their own ideas and priorities, resulting in shifts in corporate strategy that confused employees and scared off investors. In 2007, Jerry Yang returned as CEO a move that was met with optimism but ultimately proved short-lived, Under his leadership, Yahoo missed opportunities, including the failure to acquire Google. Following Yang's departure, a string of CEOs took the helm, including Carol Bartz, Scott Thompson, and Ross Levinsohn. None were able to halt Yahoo's decline, and the company struggled to compete with Google in search and advertising. Marissa Mayer's appointment in 2012 was perhaps the most high-profile attempt to revive Yahoo's fortunes. Mayer, a former Google executive, was expected to bring fresh ideas and innovation to the company - but ultimately failed to deliver. *Marissa Mayer in 2010 while she worked at Google*. While Mayer was able to embark on multiple major initiatives, such as the acquisition of Tumblr, her tenure was marked by slow progress and a failure to reverse Yahoo's downward spiral.(41)



Marissa Mayer in 2010 while she worked at Google.

8. Missed Opportunities:

8.1. Google's offer for acquisition

In 2002, Yahoo had the chance to buy Google for \$1 billion, but executives dragged their feet; by the time they decided to pursue the offer, Google's price had soared to \$3 billion. In the late '90s, search was only generating six percent of Yahoo's income stream, and the company believed it was not worth improving.(42)

8.2. Not pursuing Facebook

Yahoo attempted to purchase Facebook back in 2006 for \$1 billion, but the initial offer was refused by CEO Mark Zuckerberg. Reports indicate that the board of directors would have forced Zuckerberg to accept an offer of \$1.1 billion, yet Yahoo executives would not agree to the increased bid.(43)

8.3. Failure to merge gracefully with Microsoft

Although Microsoft's \$44.6 billion takeover bid was rejected in 2008, Yahoo ended up signing a deal with Microsoft the very next year to use the Bing search engine after failing to develop an effective search engine of their own. This was likely the result of a team of lower quality programmers, as Yahoo was known not to place emphasis on the quality of its programming staff in the way Microsoft, Facebook and others have.(44)

Dictionary

Subprime Mortgage Meltdown - The subprime mortgage meltdown was a big financial problem in the late 2000s. It happened because people who couldn't really afford to buy houses were given loans, and when they couldn't pay them back, it caused a big crisis that hurt the economy. It was a result of bad lending practices and a housing market crash.

Pitch - to present an idea for a business.

Tranches - Tranches are divisions or segments within a mortgage-backed security, each with a different level of risk and return.

Default Mortgages - Default mortgages are loans that borrowers cannot repay, leading to a loss for investors in mortgage-backed securities.

Mezzanine Layer - The mezzanine layer is an intermediate level within a mortgage-backed security, situated between the safest and riskiest tranches. It contains mortgages with medium levels of risk and is a critical area where problems can arise, as even a few defaults here can destabilize the entire security.

Short - betting against something. In finance and investing, "short" can mean taking a position that profits from the decline in the value of an asset, such as stocks or currencies.

Adjustable Interest Rates - an interest rate on a financial product, such as a loan or a mortgage, that is not fixed and can change over time.

Credit Default Swaps - Credit Default Swaps (CDS) are like insurance for loans or bonds. If the borrower can't repay, the CDS pays the lender. It's a way to reduce the risk of not getting your money back when you lend it to someone.

Bankruptcy - Bankruptcy is a legal status of an individual or organization that cannot repay their debts, and it involves a formal process to resolve or eliminate those debts.

Brokerage Firm - A brokerage firm is a financial institution or company that facilitates the buying and selling of various financial assets, such as stocks, bonds, and other securities, on behalf of clients.

Investment Bank - An investment bank is a financial institution that provides services such as underwriting, mergers and acquisitions, trading, and asset management for corporations, governments, and other large entities.

Stocks - Stocks, also known as shares or equities, represent ownership in a company. When you own a stock, you own a portion of the company and may have the right to participate in its profits and decision-making.

Evolution - Evolution refers to the gradual development or change of something over time. In the financial context, it can refer to the transformation of financial institutions, markets, or products.

Asset - An asset is a resource with economic value that an individual, company, or organization owns or controls with the expectation that it will provide future benefit.

Market Shares Value - Market share value refers to the worth of a company's portion of the total market for a specific product or service. It's a measure of the company's relative size or significance in that market.

Money Laundering - Money laundering is the process of disguising the origins of illegally obtained money, typically by passing it through a complex sequence of banking transfers or commercial transactions to make it appear legitimate.

Book Value - Book value is the net value of a company's assets minus its liabilities, as recorded on its balance sheet. It represents the value of a company's assets if all its debts were paid off.

Reserves - Reserves, in a financial context, refer to funds or assets set aside for specific purposes or to cover potential future losses or contingencies.

Solvent - Solvent means that an individual, company, or organization is financially stable and able to meet its financial obligations, including paying off debts when they are due.

References

- (1) - [Movie] The Big Short directed by Adam McKay: https://youtu.be/psCF_AHuLM4?si=7wlcpknyh9daPuEq
- (2) - [Image] Lewie Ranieri wants to fix the mortgage mess by Shawn Tully: https://money.cnn.com/2009/12/08/real_estate/lewie_ranieri_mortgages.fortune/index.htm
- (3) - Mortgage-Backed Securities (MBS) Definition: Types of Investment by Julia Kagan: <https://www.investopedia.com/terms/m/mbs.asp>
- (4) - Why Do MBS (Mortgage Backed Securities) Still Exist? by Sean Ross: <https://www.investopedia.com/ask/answers/040915/why-do-mbs-mortgagebacked-securities-still-exist-if-they-created-so-much-trouble-2008.asp>
- (5) - How it Happened - The 2008 Financial Crisis: Crash Course Economics #12 by CrashCourse: <https://youtu.be/GPOv72Awo68?si=DGuNAR49FnKCzdmF>
- (6) - Real Wall Street Expert Reviews The Big Short (Jenga Scene) by Financial Edge Training: <https://www.youtube.com/watch?v=o7z5Wbq33yE>
- (7) - Trader who predicted 2008 financial crisis bets \$1.6bn on stock market crash by end of 2023 by William Mata: <https://www.independent.co.uk/news/world/americas/michael-burly-stock-market-big-short-b2395947.html>
- (8) - Crisis in Prosperous Societies: Are They Inevitable? by Yoram Kirsh: <https://deliverypdf.ssrn.com/delivery.php?ID=766114027119125064126093018084109024041036054041060067089083110124119125127080030065030103098020118>
- (9) - [Book] Research handbook on Disability & Entrepreneurship by Shumaila Yousufzai, Wilson Ng, Shandana Sheikh, & Thomas Coogan: https://books.google.com.pk/books?hl=en&lr=&id=hW-VEAAQBAJ&oi=fnd&pg=PA59&dq=Dr.+Michael+Burly&ots=vve-mLC3RW&sig=7OWQ-X6zaAOk64MAI3D_DAOEU4&redir_esc=y#v=onepage&q=Dr.+Michael+Burly&f=false
- (10) - 15 years ago Lehman Brothers became the 'only true icon to fall in a tsunami' that rocked the global economy by Bob Bryan: <https://www.google.com/amp/s/www.businessinsider.com/lehman-brothers-bankruptcy-anniversary-icon-in-a-tsunami-2016-9%3Famp>

- (11) - Largest bankruptcies in the United States as of July 2023, by assets at time of bankruptcy by Statista Research Department: https://www.statista.com/statistics/1096794/largest-bankruptcies-usa-by-assets/?gclid=Cj0KCQjwm66pBhDQARIsALIR2zC8TJRQHMDQz6AYQgW_TuR2gkQP01j12rqg1KCBsTSYShXp9scVn6laAq5eEALw__w
- (12) - The Collapse of Lehman Brothers: A Case Study by Nick Lioudis: <https://www.investopedia.com/articles/economics/09/lehman-brothers-collapse.asp>
- (13) - The Collapse of Lehman Brothers - A Simple Overview by Company Man: https://youtu.be/BnDbdQa_r38?si=UfyoSkd3yGM2WlXX
- (14) - Lehman Brothers by Wikipedia: [https://en.wikipedia.org/wiki/Lehman_Brothers#:~:text=Merger%20with%20American%20Express%20\(1984%E2%80%93931994\)Shearson%20Lehman%2FAmerican&text=Shearson%2FAmerican%20Express%2C%20an%20American,became%20Shearson](https://en.wikipedia.org/wiki/Lehman_Brothers#:~:text=Merger%20with%20American%20Express%20(1984%E2%80%93931994)Shearson%20Lehman%2FAmerican&text=Shearson%2FAmerican%20Express%2C%20an%20American,became%20Shearson)
- (15) - The impact of Lehman Brothers on Romanian banks listed on BVB by Ciupac-Ulici Maria-Luneta: https://www.researchgate.net/figure/Figure-no1-Evolution-of-Lehman-Brothers-stock-during-June-2004-October-2010_fig5_241751939
- (16) - Deutsche Bank Slides as European Bank Worries Persist by Bloomberg Television: https://youtu.be/fPb_M0LMuvQ?si=wWM6Y_1WgN0yA6Cc
- (17) - Inspired by Deutsche Bank Death Spiral, European Banks Sink to Dec 24, 2018 Level – First Seen in 1995 by Wolf Richter: <https://wolfstreet.com/2019/06/15/led-by-deutsche-bank-death-spiral-european-banks-sink-to-dec-24-level/>
- (18) - Deutsche Bank fined record \$2.5 billion over rate rigging by Kirsten Ridley: <https://www.reuters.com/article/us-deutschebank-libor-settlement-idUSKBN0NE12U20150423>
- (19) - Deutsche Bank to pay \$258 million to settle U.S. sanctions case - NYDFS by Suzanne Bariyn: <https://www.reuters.com/article/us-deutsche-bank-penalty-idUSKCN0ST2J720151104>
- (20) - Deutsche Bank agrees to \$7.2 billion mortgage settlement with U.S. by Karen Freifeld, Arno Schuetze, and Kathrin Jones: <https://www.reuters.com/article/us-deutsche-bank-mortgages-settlement-idUSKBN14C041>
- (21) - Deutsche Bank fined for \$10 billion Russian money-laundering scheme by Jethro Mullen: <https://money.cnn.com/2017/01/31/investing/deutsche-bank-us-fine-russia-money-laundering/index.html>
- (22) - Deutsche Bank's Violation Tracker Current Parent Company Summary by Violation Tracker: <https://violationtracker.goodjobsfirst.org/parent/deutsche-bank>
- (23) - Trump and Deutsche Bank: It's Complicated by Heather Vogell and Andrea Bernstein: <https://www.wnycstudios.org/podcasts/trumpinc/episodes/trump-inc-trump-deutsche-bank-its-complicated>
- (24) - Deutsche Bank will exit global equities business and slash 18,000 jobs in sweeping overhaul by Spencer Kimball: <https://www.cnbc.com/2019/07/07/deutsche-bank-will-exit-its-global-equities-business-and-scale-back-investment-bank.html>
- (25) - Mobile Money by Alan Moore: <https://communities-dominate.blogs.com/brands/mobile-money/>
- (26) - How is Nokia Even Still Alive: <https://www.youtube.com/watch?v=PL83PinTIkI>
- (27) - Nokia net sales worldwide from 1999 to 2022 by Thomas Alsop: <https://www.statista.com/statistics/267819/nokias-net-sales-since-1999/>
- (28) - Revenue history for Nokia from 2001 to 2023 by CompaniesMarketCap: <https://companiesmarketcap.com/nokia/revenue/>
- (29) - The Strategic Decisions That Caused Nokia's Failure by Yvez L. Doz: <https://knowledge.insead.edu/strategy/strategic-decisions-caused-nokias-failure>
- (30) - Nokia CEO Stephen Elop to rejoin Microsoft by Nic Healey: <https://www.cnet.com/tech/mobile/nokia-ceo-stephen-elop-to-rejoin-microsoft/>
- (31) - How Stephen Elop led to the fall of Nokia? by Pulkit K.: <https://www.linkedin.com/pulse/how-stephen-elop-led-fall-nokia-pulkit-khare>
- (32) - TIMELINE-From paper mills to 5G: The many lives of Nokia by Reuters Staff: <https://www.reuters.com/article/idUSL5N2AO2IE>
- (33) - YAHOO! FIASCO: A SINKING VESSEL NEEDS NO NAVIGATION by Dilipraj, E.: https://capsindia.org/wp-content/uploads/2021/10/CAPS_Infocus_DR_22.pdf
- (34) - YAHOO! FIASCO: A SINKING VESSEL NEEDS NO NAVIGATION by Dilipraj, E.: https://capsindia.org/wp-content/uploads/2021/10/CAPS_Infocus_DR_22.pdf

- (35) - **Revenue of Yahoo from 2004 to 2016** by **Tiago Bianchi**: <https://www.statista.com/statistics/266253/yahoos-annual-gaap-revenue/>
- (36) - **How Google Came To Dominate Search And What The Future Holds** by **Adam Heitzman**: <https://www.forbes.com/sites/forbesagencycouncil/2017/06/05/how-google-came-to-dominate-search-and-what-the-future-holds/?sh=2973c01c3872>
- (37) - **What Happened to Yahoo! in Six Points** by **University Of Maryland**: <https://onlinebusiness.umd.edu/blog/what-happened-to-yahoo-in-6-points/>
- (38) - **4 Biggest Reasons Why Yahoo! Failed** by **Shrishti Mathur**: <https://www.feedough.com/why-yahoo-failed/>
- (39) - **YAHOO! FIASCO: A SINKING VESSEL NEEDS NO NAVIGATION** by **Dilipraj, E.**: https://capsindia.org/wp-content/uploads/2021/10/CAPS_Infocus_DR_22.pdf
- (40) - **What Happened to Yahoo! in Six Points** by **University Of Maryland**: <https://onlinebusiness.umd.edu/blog/what-happened-to-yahoo-in-6-points/>
- (41) - **Why did Yahoo Fail? The Rise and Fall of a Dot-Com Tech Giant** by **Ellis Stewart**: <https://em360tech.com/tech-article/why-did-yahoo-fail>
- (42) - **What Happened to Yahoo! in Six Points** by **University Of Maryland**: <https://onlinebusiness.umd.edu/blog/what-happened-to-yahoo-in-6-points/>
- (43) - **What Happened to Yahoo! in Six Points** by **University Of Maryland**: <https://onlinebusiness.umd.edu/blog/what-happened-to-yahoo-in-6-points/>
- (44) - **What Happened to Yahoo! in Six Points** by **University Of Maryland**: <https://onlinebusiness.umd.edu/blog/what-happened-to-yahoo-in-6-points/>